# Endothermic & Exothermic Reactions

**Exam code:** 0620 & 0971

Download PDF

**Written by:** Alexandra Brennan
**Reviewed by:** Caroline Carroll
Updated on 22 December 2025

SaveMyExams

Did this video help you? Yes No

# Exothermic & endothermic reactions

## Heat exchange in reactions

- Chemical reactions occur so that elements can achieve a more stable energy state by gaining a full outer shell of electrons
  - This is done by **chemical bonding**
- This process involves the **transfer of thermal energy** into and out of reaction mixtures
- The terms used to describe this are:
  - **System:** the reacting chemicals
  - **Surroundings:** anything other than the chemicals reacting
- The energy within the system comes from the **chemical bonds** themselves which could be considered as tiny stores of chemical energy

## Exothermic reactions

- In exothermic reactions, thermal energy is **transferred** from the system to the surroundings
  - The energy of the system decreases, which means that the energy change is **negative**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

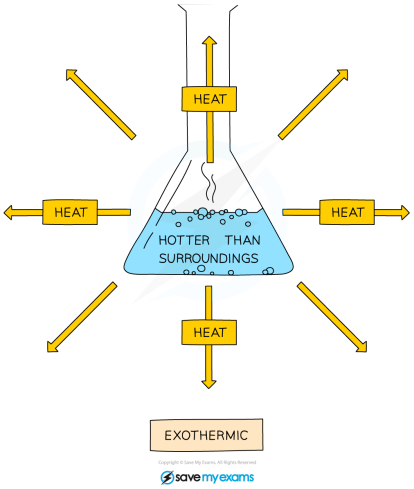



***Diagram showing the transfer of heat energy outwards from an exothermic reaction***

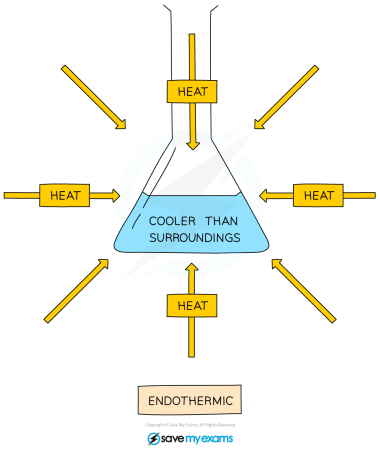
- Typical examples of exothermic reactions include:
  - **Combustion**
  - **Oxidation**
  - **Neutralisation**

- **Hand warmers** used in the wintertime are based on the release of heat from an exothermic reaction
- **Self-heating cans** of food and drinks such as coffee and hot chocolate also use exothermic reactions in the bases of the containers

## Endothermic reactions

- In endothermic reactions, thermal energy is **transferred** from the surroundings to the system
  - The energy of the system increases, which means that the energy change is **positive**
  - The temperature of the surroundings decreases because thermal energy is **taken in / absorbed**

- The overall transfer is from the surroundings to the system



- Endothermic reactions are less common than exothermic reactions
- Typical examples of endothermic reactions include:
  - Electrolysis
  - Thermal decomposition
  - The first stages of photosynthesis

- **Cold packs** for sports injuries are based on endothermic reactions, designed to take heat away from a recently injured area to prevent swelling

### Worked Example

A student was investigating the temperature change for four different chemical reactions. The table shows the chemicals that the student combined for each reaction along with the initial and final temperatures of the reaction.

| Experiment | Chemicals | | Initial temperature (°C) | temp |
|---|---|---|---|---|
| **1** | 10 $cm^3$ NaOH | 10 $cm^3$ HCl | 19 | |
| **2** | 10 $cm^3$ $NaHCO_3$ | 2 g citric acid | 20 | |
| **3** | 10 $cm^3$ $CuSO_4$ | 0.5 g Mg powder | 20 | |
| **4** | 10 $cm^3$ $H_2SO_4$ | 3 cm Mg ribbon | 19 | |

Explain whether each reaction is endothermic or exothermic.

**Answers:**

- Reactions 1, 3 and 4 are exothermic reactions because they show a temperature increase
- Reaction 2 is an endothermic reaction because it shows a temperature decrease

## Reaction pathway diagrams

- **Reaction pathway diagrams** are graphical representations of the relative energies of the reactants and products in chemical reactions
- On a reaction pathway diagram:
  - Progress of the reaction is shown on the x-axis
  - Energy is shown on the y-axis
  - The difference in height between the energy of reactants and products is the **overall energy**

***Reaction pathway diagram of an exothermic reaction and an endothermic reaction***

- In exothermic reactions:
  - Energy is given out to the surroundings
  - The energy of the products will therefore be **lower** than the energy of the reactants
  - The overall energy change is **negative**
  - This is represented on the reaction profile with a downwards-arrow as the energy of the products is lower than the reactants
- In endothermic reactions:
  - Energy is taken in from the surroundings
  - The energy of the products will be **higher** than the energy of the reactants
  - The overall energy change is **positive**
  - This is represented on the reaction profile with an upwards-arrow as the energy of the products is higher than the reactants

## Examiner Tips and Tricks

- To help you remember whether a chemical system is exothermic or endothermic:
  - In **EX**othermic reactions heat **Ex**its the system and in **EN**dothermic reactions heat **EN**ters the system.
  - Exothermic reactions always give off heat and they feel hot
  - Endothermic reactions always take heat in and they feel cold.
- Core candidates will be expected to interpret reaction pathway diagrams
- Extended candidates will be expected to draw and interpret reaction pathway diagrams

Teacher only

## New! Exclusive Teacher-Only Questions

Fresh, expert-written questions

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

**Resources**

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

**Company**

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

**Dashboard**

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

**Subjects**

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects